

Hammad Ahmad

AI / Machine Learning Engineer — LLMs · RAG · Knowledge Graphs

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

PROFILE

AI/ML engineer with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication. I build RAG systems pairing vector embeddings with knowledge graphs, plus predictive ML models on real-world data, and work comfortably across the stack — from retrieval design to backend APIs and deployment.

TECHNICAL SKILLS

Machine Learning & NLP : PyTorch, TensorFlow, scikit-learn, LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, SHAP, MLflow

LLM & Retrieval : Vector embeddings, semantic search, cross-encoder re-ranking, RAGAS evaluation, Neo4j knowledge graphs

Engineering : Python, C++, SQL, FastAPI, Flask, REST APIs, React

Data & Infrastructure : PostgreSQL, Neo4j, pandas, NumPy, Docker, Git, Linux

EXPERIENCE

Outlyst Oct 2025 – Mar 2026

AI / Machine Learning Engineer Leeds, UK (Remote)

- Contributed to the development of an AI voice-agent system built with Retell AI and FastAPI, working on backend dialogue-flow logic and workflow automation.
- Worked with asynchronous backend services and lead-tracking automation for an AI-driven outbound communication tool.
- Gained hands-on experience integrating LLM-based dialogue systems with production backend infrastructure.

University of Bradford Jan 2025 – Sep 2025

Research Assistant — Machine Learning & LLMs Bradford, UK

- Built FinLaw-UK, a RAG system pairing an open-source LLM (Mistral 7B) with a Neo4j knowledge graph for reliable UK financial-regulation question answering.
- Improved retrieval quality by fine-tuning Sentence Transformer embeddings and adding cross-encoder re-ranking.
- Benchmarked answer faithfulness and relevance using the automated RAGAS evaluation framework.

COMSATS University Islamabad Jul 2023 – Jul 2024

Research Assistant — Data Science Islamabad, Pakistan

- Developed ensemble predictive models (Random Forest, Gradient Boosting) on clinical datasets, reaching ~93% classification accuracy under stratified cross-validation.
- Applied SHAP-based interpretability for model transparency and deployed models via REST APIs.

PROJECTS

Jobzyl — Multi-Platform Job Aggregator jobzyl.com

Full-Stack Architecture, Data Pipelines, Responsive UI

- Built a full-stack platform that ingests real-time job listings from multiple sources (LinkedIn, Indeed, Google Jobs) into a single searchable portal.
- Designed the data-parsing pipelines and responsive UI; live and actively maintained.

EDUCATION

University of Bradford Sep 2024 – Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics — Merit Bradford, UK

- **Dissertation**: FinLaw-UK — A Graph-Augmented Retrieval Chatbot for Reliable UK Financial Regulation.

COMSATS University Islamabad Sep 2020 – Jul 2024

BS, Bioinformatics Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1

LANGUAGES

English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)